

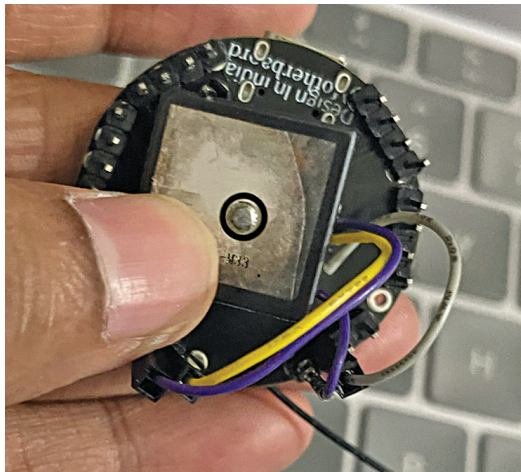


Fig. 6: Module fixing on the bottom of IndusBoard

According to the L86 GPS module datasheet, the module can be powered from a range of 2.8V to 4V while consuming around 100mA in working mode. The four output pins on IndusBoard can handle the current and power. So, pins 3.3V and GND of the IndusBoard are used for GPS powering.

The GPS module has the RTC backup power pin as per the datasheet. This helps the RTS to keep

track of time when the main power is out. Here, any 3.3V coin cell can be attached. In this design, a 3.3V source is used for RTC power backup as well. Fig. 4 shows the GPS SMD pinouts details.

The GSM module SIM8001 or SIM808 or any other equivalent is added to the IndusBoard. RX and TX pins of SIM8001 GSM module are connected to pins 41 and 42 of the IndusBoard, respectively. Pins 3.3V and GND power supply of the board are also connected to power the GSM module.

If the network status, messages received, location latitudes, and longitudes are desired, the SSD1306 OLED display needs to be added to the device as shown in Fig. 2. However, this will increase the cost of the material used and power consumption of the device. It is not included in the software, but in the design, the display is provided in the circuit as well.

Install the AsyncSMS, HardwareSerial, TinyGPS ++, and SIM libraries into IndusBoard. Then select the baud rate of both GPS

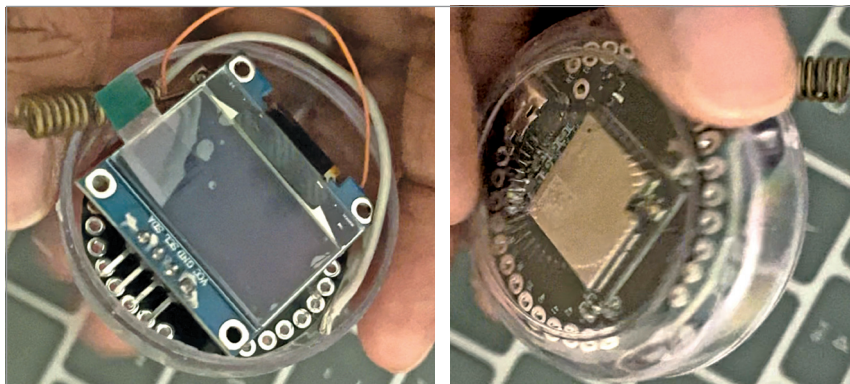


Fig. 7: GPS tracker using IndusBoard and L86 GPS module



Fig. 8: GPS tracker ring

